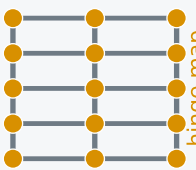
Recoverability is a state-space crossing

The hinge map changes the path by which a frame reaches the barrier.

1 Current nonlinear state

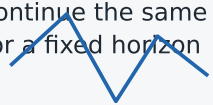
frame + hinge map
state: u, v, S, F_y



released at
the current state

2 Remove future input

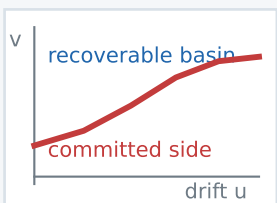
ground acceleration -> 0
continue the same model
for a fixed horizon



zero-input continuation

3 Barrier in (u, v)

static saddle = v = 0 point
of the separatrix



inside

4a Recoverable

state returns to the
elastic neighborhood

crossed

4b Committed

state escapes after
input removal

Hinge topology and capacity allocation

beam ends + bases
local hinge map

→

column ends added
admissible path

→

multi-bay budget
capacity split

barrier shifts

Mechanism schematic; quantitative evidence is provided by the phase-plane and frame census figures.